MLLP MINI PROJECT · TASK 13

Keyword Extraction from Scanned Academic Notices

OCR + TF-IDF + KeyBERT, evaluated on 12 hand-labelled college notices

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Problem: scanned notices can't be searched

The problem

Colleges post circulars as scanned images. A scan is just pixels: it cannot be searched, tagged or filtered, so students must open and read every notice to find the one they need.

Our objectives

1. Scan → text with OpenCV + Tesseract OCR
2. Extract keywords with TF-IDF and KeyBERT
3. Score both against hand-labelled keywords
4. Ship a usable web demo (Gradio)

Goal: turn every scanned notice into a short list of searchable keywords, and measure how good that list is.

Dataset: 12 notices, 4 categories

3

Exam

timetable, results, practicals

3

Fee

tuition, hostel, scholarship

3

Holiday

Christmas, winter, Guru Nanak Jayanti

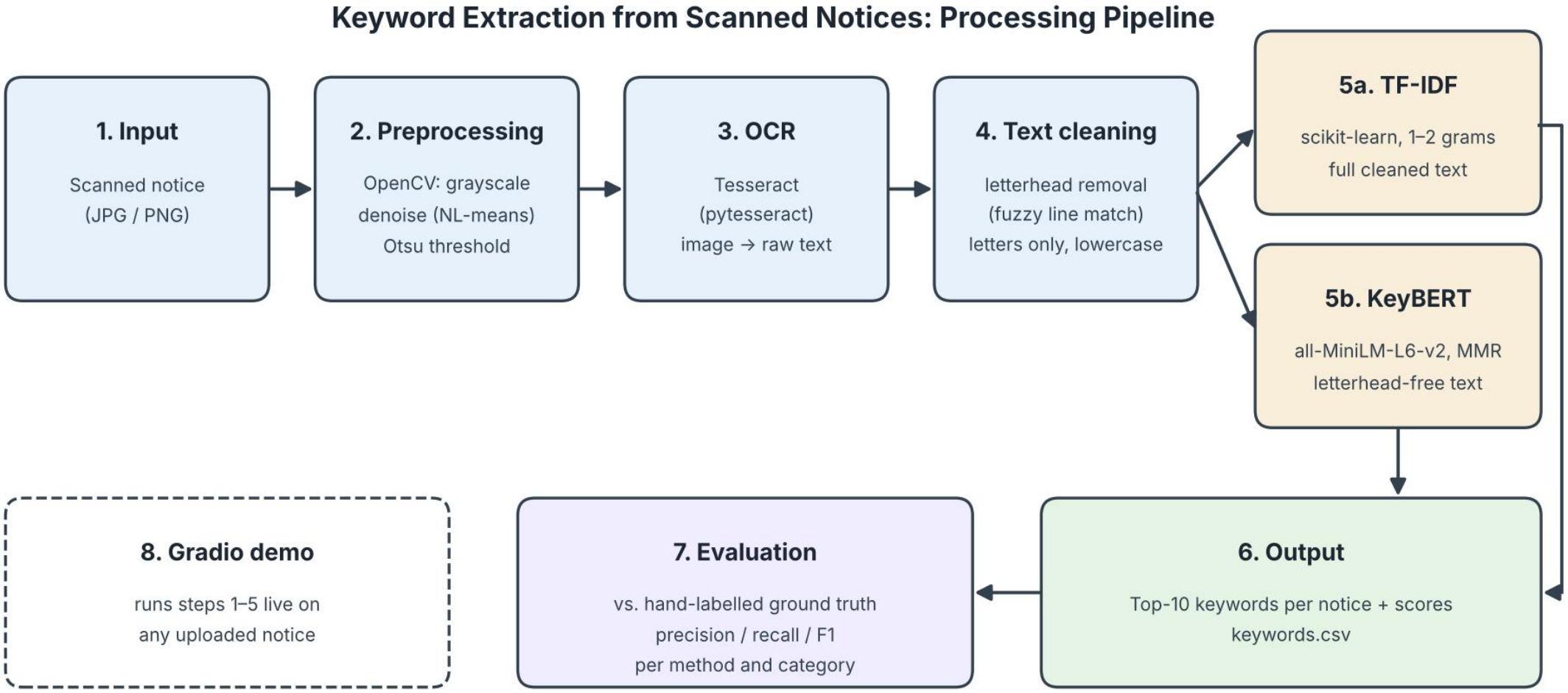
3

Event

tech fest, guest lecture, sports meet

Each notice has a letterhead, reference number, title, body and signatory. For every notice we hand-labelled **6–7 ground-truth keywords**, e.g. *end semester examination, hall ticket, exam fee*.

Pipeline: scan to keywords in six stages



Models and tools

Stage	Model / library	What it does
Preprocessing	OpenCV	Grayscale, NL-means denoise, Otsu threshold
OCR	Tesseract (pytesseract)	Image → raw text
Statistical keywords	scikit-learn TF-IDF (1–2 grams)	Frequent in this notice, rare in others
Semantic keywords	KeyBERT + MMR (diversity 0.3)	Phrases closest in meaning to the notice
Embeddings	sentence-transformers/all-MiniLM-L6-v2	384-dim sentence vectors (Hugging Face)
Demo	Gradio	Upload a notice, see the keywords

Google Colab, CPU only. No training: every model is used pretrained.

Key finding: KeyBERT was reading the letterhead

Before: full text

eastern institute · bhubaneswar odisha · **examination timetable** · institute technology · eitm coe …

F1 0.261

After: letterhead removed + MMR

examination timetable · semester examinations · examination hall · end semester …

F1 0.571

Lines in ≥50% of notices are letterhead. We remove them with **fuzzy matching (85%)**, so OCR slips like "Copy ta" for "Copy to" are caught too.

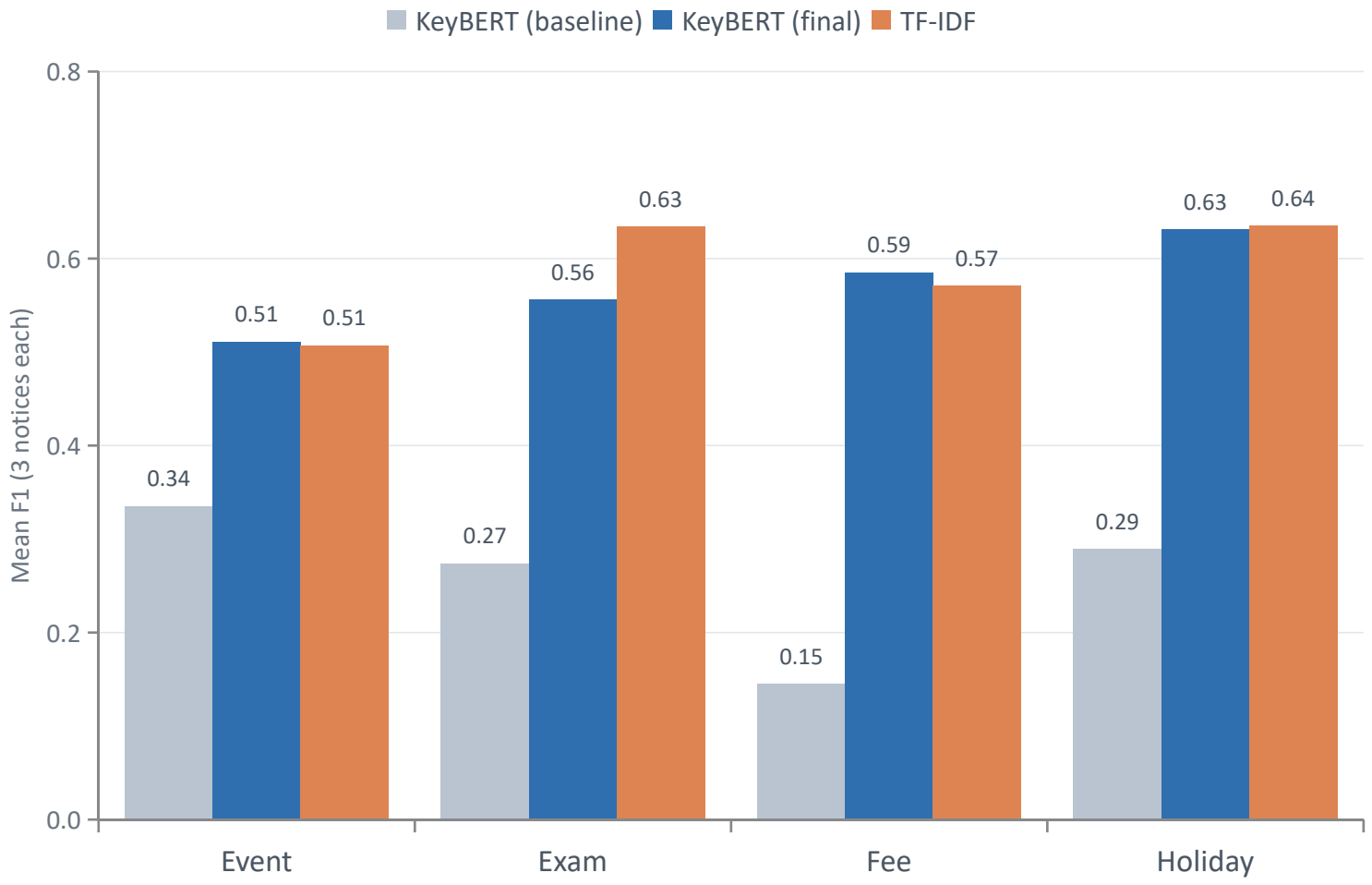
Example: notice1.jpg

Results: precise TF-IDF vs. broader KeyBERT

Method	Precision	Recall	F1
TF-IDF	**0.671**	0.535	**0.587**
KeyBERT (baseline)	0.238	0.292	0.261
KeyBERT (final)	0.502	**0.676**	**0.571**

Metric: phrases split into words (stopwords and digits removed), compared with the ground truth. Mean over 12 notices.

By category: TF-IDF wins on exams



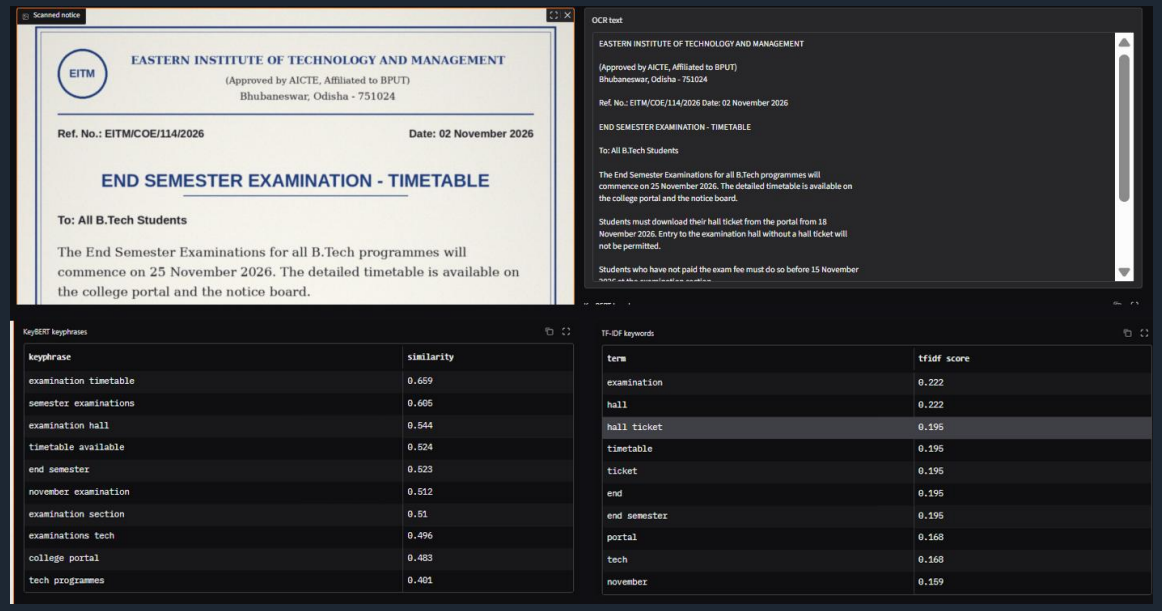
Exam: TF-IDF 0.63 vs KeyBERT 0.56

Fee, event: KeyBERT slightly ahead

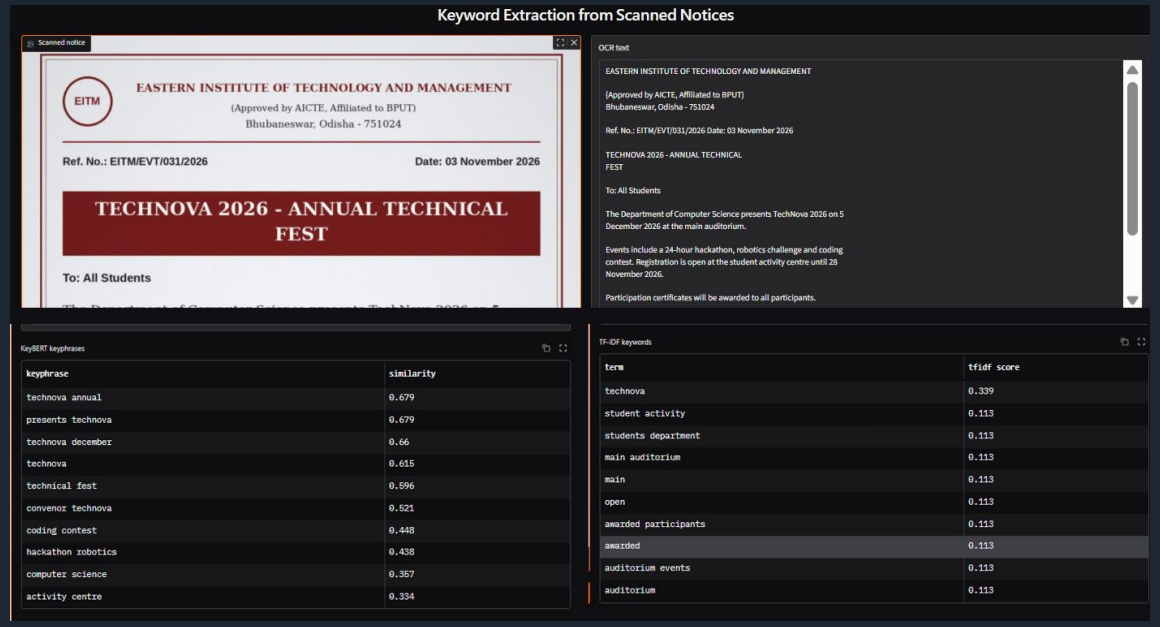
Holiday: a tie

KeyBERT wins on 5 of 12 notices; per-notice F1 ranges 0.30–0.73.

Demo: upload a notice, get its keywords



Exam notice: KeyBERT gives clean phrases; TF-IDF splits "hall ticket" into fragments.



Event notice: KeyBERT repeats "technova" 5×; TF-IDF ties 9 terms at 0.113.

Gradio app with a public gradio.live link

Limitations and future scope

Limitations

- Only 12 notices from one institute
- Clean digital scans, not phone photos
- Dates and amounts are stripped out
- Word-level metric is lenient
- English only; no tables or columns

Future scope

- Extract dates and fees with NER / regex
- Hybrid: TF-IDF candidates, KeyBERT ranking
- Stronger MMR diversity to stop repeats
- 100+ real notices, with a test split
- Hindi and Odia OCR + multilingual model

Biggest next step: keep dates and fee amounts, the facts students search for most.

Team and contributions

Name	Roll number	Contribution
Karan Buragohain	2304024	Coding, Implementation,Github and Data curation
Bikash Pradhan	2304020	Model Traning, Datacuration , Report
Ayushmaan Nath	2304019	GitHub, Matplotlib, PPT , Presentation

Code, data and report: https://github.com/i-karanburagohain/keyword-extraction-scanned-doc
Video of Presentation: https://drive.google.com/file/d/1IqWBYf7Wyr79DL2a1_Tb96jweYh3c1wC/view?usp=sharing

Thank you